

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – Street Vendor Digitalization Agent

Domain of Project – Social & Economic Development / MSME

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Problem Statement

Problem Statement : Street Vendor Digitalization Agent

The Challenge

The key challenge is to design and implement an **AI-powered assistant** that can:

- Simplify onboarding for vendors with minimal technical knowledge
- Provide multilingual support for diverse communities
- Integrate financial tools like UPI seamlessly
- Retrieve and explain government schemes in a vendor-friendly way
- Ensure scalability and affordability using cloud-based services

The Objective

The **Street Vendor Digitalization Agent** aims to:

1. **Empower vendors** with easy-to-use digital profiles and payment solutions.
2. **Enhance visibility** through local SEO and personalized marketing guidance.
1. **Bridge knowledge gaps** by retrieving and explaining relevant MSME schemes.
2. **Promote inclusivity** with multilingual support and simple interfaces.
3. **Leverage IBM Granite & Cloud Lite** to deliver scalable, secure, and cost-effective AI services.

Technology Used



Frontend
React.js / Angular,
HTML5, CSS3,
TailwindCSS / Bootstrap



Backend
Node.js, Express.js,
IBM Granite
(Watsonx.ai)



Database
MongoDB Atlas /
Local MongoDB



Authentication &
Security
JWT, CORS, HTTPS



Cloud & Deployment
IBM Cloud Lite,
Vercel / Netlify, Render /
Heroku



Development & Tools
Git & GitHub,
Postman, VS Code, npm
/ yarn

Proposed Solution -

The proposed solution is a **multi-agent AI system** built using **IBM Granite Models, Watsonx.ai, and IBM Cloud Lite services** to empower street vendors with digital visibility, financial inclusion, and customer engagement. The system addresses challenges in vendor onboarding, UPI adoption, local SEO, and access to government schemes through an intelligent, modular architecture. At its core, the solution leverages a **Retrieval-Augmented Generation (RAG) pipeline** to fetch real-time data from trusted sources such as government MSME portals, fintech APIs, and commerce platforms. This data is embedded and stored in a vector database, enabling the system to retrieve and contextualize policies, pricing tips, and promotional strategies using Granite models.

Key Agents

- **Business Profile Agent** → Generates vendor profiles with name, location, products, and multilingual descriptions for online listings.
- **UPI Setup Agent** → Guides vendors step-by-step in creating QR codes, linking bank accounts, and enabling digital payments.
- **Local SEO Agent** → Suggests keywords, geo-tags, and listing platforms to improve visibility in hyperlocal searches.
- **Customer Engagement Agent** → Provides promotional materials, loyalty tips, and language-specific communication strategies.

User Interface

- A **web dashboard (React/Angular)** for vendors to onboard easily.
- Mobile-friendly forms for profile creation and payment setup.
- Visual analytics showing customer reach, payment adoption, and engagement metrics.

Orchestration & AI

- **LangFlow / Orchestration Layer** → Manages agent workflows.
- **Watsonx.ai** → Ensures scalable model deployment, embeddings, and AI governance.
- **Granite Models** → Power reasoning, summarization, and personalized recommendations.

Langflow component Used -

Input Component (Forms / Chat Interface)

- Captures vendor details (name, business type, location, products, language).
- Supports multilingual text input for inclusivity.

RAG Component (Retrieval-Augmented Generation)

- Connects to vector database storing MSME schemes, UPI guides, and local SEO tips.
- Retrieves relevant documents and passes them as context to Granite models.

Granite Model Component (LLM Node)

- Powers reasoning, summarization, and personalized recommendations.
- Used for generating vendor profiles, onboarding instructions, and customer engagement tips.

Agent Router Component

- Directs queries to the correct agent (Profile Agent, UPI Agent, SEO Agent, Engagement Agent).
- Ensures modularity and scalability.

Database Connector (MongoDB)

- Stores vendor profiles, onboarding progress, and customer interaction logs.
- Enables persistence and analytics.

Output Component (Dashboard / API Response)

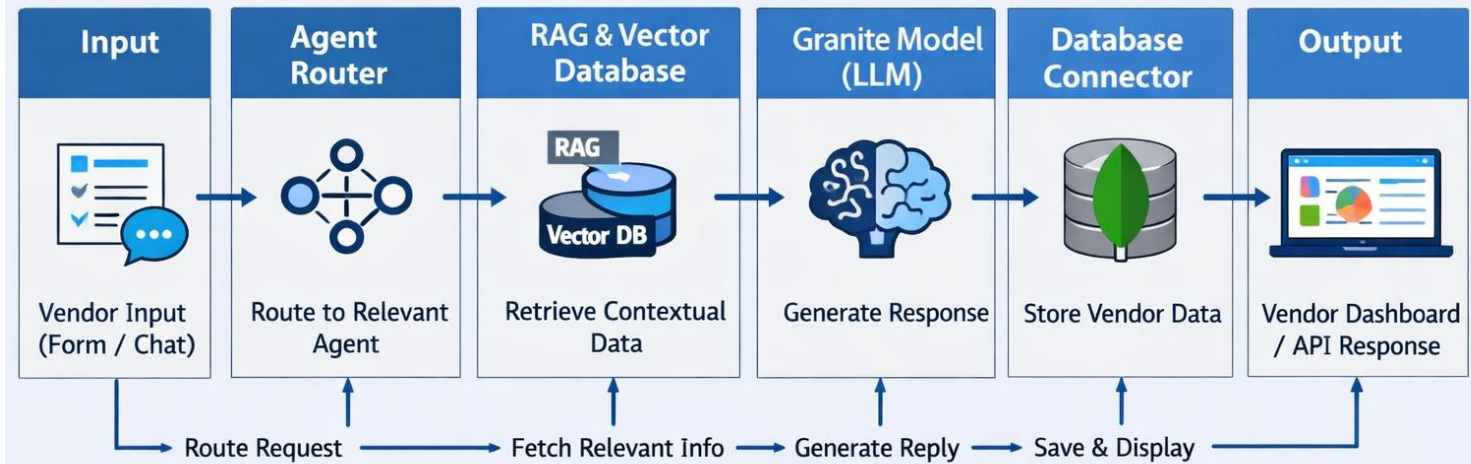
- Displays onboarding success, payment setup status, and visibility metrics.
- Provides actionable insights in a simple UI.

Specialized Agents in LangFlow

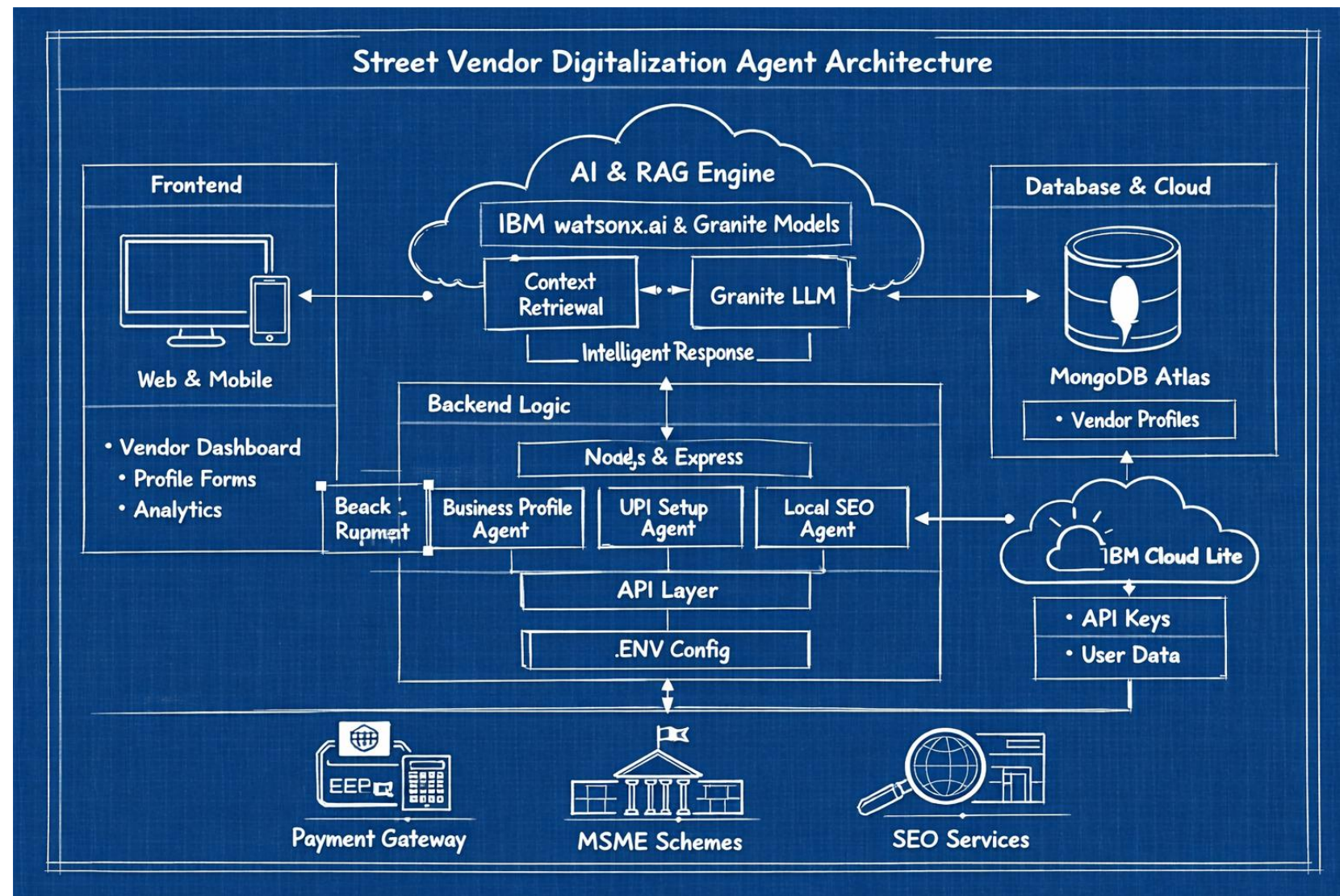
- **Business Profile Agent** → Generates multilingual vendor profiles.
- **UPI Setup Agent** → Guides vendors through digital payment onboarding.
- **Local SEO Agent** → Suggests keywords and geo-tags for visibility.
- **Customer Engagement Agent** → Provides promotional strategies and communication tips.

Langflow workflow -

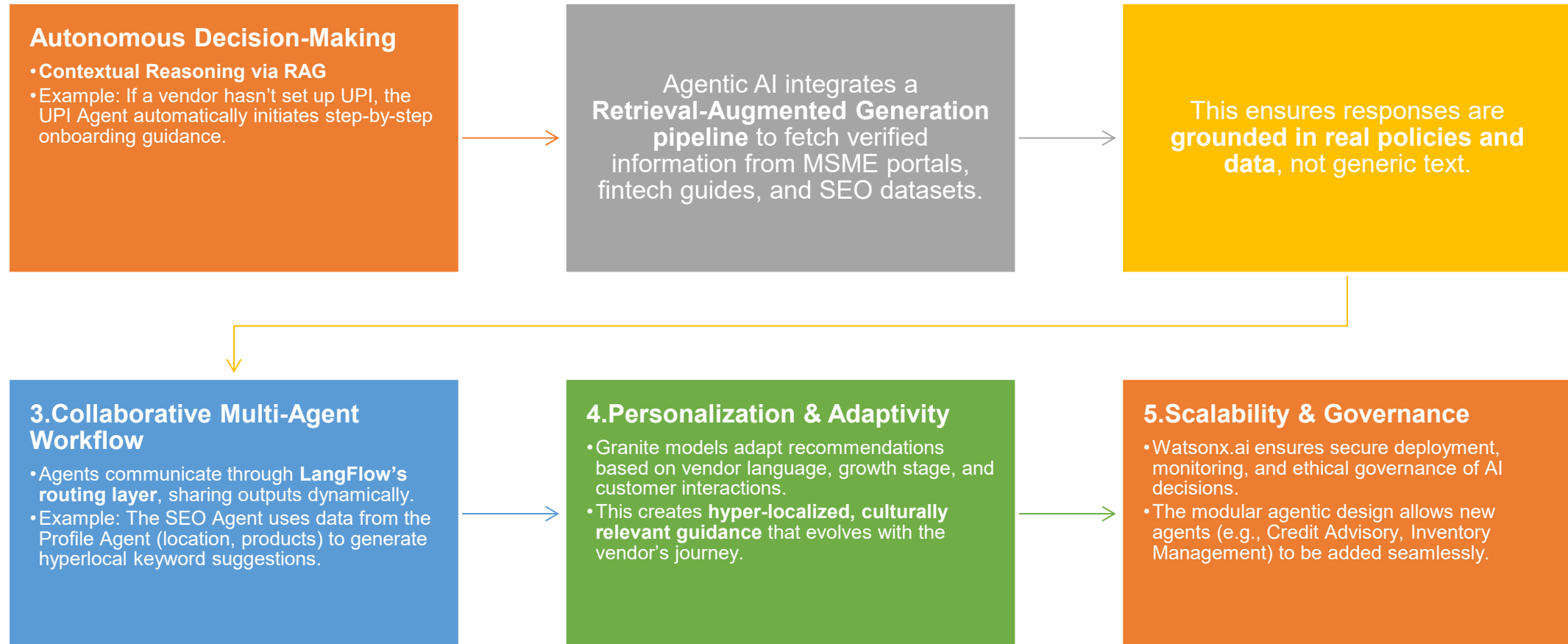
LangFlow Workflow for Street Vendor Digitization Agent



Architecture blueprint



Role of Agentic AI in the solution



Technology Used

- **AI & Orchestration** → IBM Granite Models, Watsonx.ai, LangFlow (multi-agent + RAG)
- **Backend** → Node.js, Express.js, JWT Authentication
- **Frontend** → React.js / Angular, HTML5, CSS3, TailwindCSS/Bootstrap
- **Database** → MongoDB Atlas / Local MongoDB, Vector DB (for embeddings)
- **Cloud & Deployment** → IBM Cloud Lite, Vercel/Netlify (frontend), Render/Heroku (backend)
- **Dev Tools** → Git/GitHub, Postman, VS Code, npm/yarn

Novelty and Uniqueness

- The proposed Street Vendor Digitalization Agent stands out by combining **Agentic AI, RAG, and personalization** into a single intelligent system.
- **Agentic AI Approach** → Multi-agent system (Profile, UPI, SEO, Engagement) working autonomously yet collaboratively.
- **RAG Integration** → Grounded responses using MSME schemes, fintech guides, and local SEO data.
- **Vendor-First Design** → Simple, multilingual onboarding tailored for low-tech users.
- **End-to-End Digitalization** → Combines payments, visibility, and customer engagement in one platform.
- **Scalable & Modular** → Easily extendable with new agents (credit advisory, inventory management).

Git Hub Link

- <https://github.com/Nissy-niveditha21/Street-Vendor-Digitalization-Agent>

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Future Scope

1. **Credit Advisory Agent** → Help vendors access micro-loans and government credit schemes.
2. **Inventory Management Agent** → Track stock, sales, and automate restocking alerts.
3. **Multilingual Voice Interface** → Voice-based onboarding and query handling for low-literacy users.
4. **Integration with E-Commerce Platforms** → Enable vendors to list products on marketplaces.
5. **AI-Driven Analytics** → Predict customer demand, seasonal trends, and optimize pricing.

Thank You!

Thank you for your time and interest.